

Drilling Mechanics - APWD Time Based

Schlumberger

Real Time, Recorded Mode, Composite

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

Longitude: 3° 26' 36.13" E

Block: 31/5

UWID:

Rig Name:

Rig Type:

West Hercules

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 8 1/2 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 29-Dec-2019 -- 15-Jan-2020

Log Interval: 12/29/2019 8:00:00 AM -- 1/15/2020 4:00:00 AM

Index Types: Time

Index Scales: 5 cm / 3600 secs

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

Other Services:

Direction & Inclination (Surveys)

Directional Drilling

geoVISION Bit Resistivity

VISION Resistivity

Disclaimer

THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

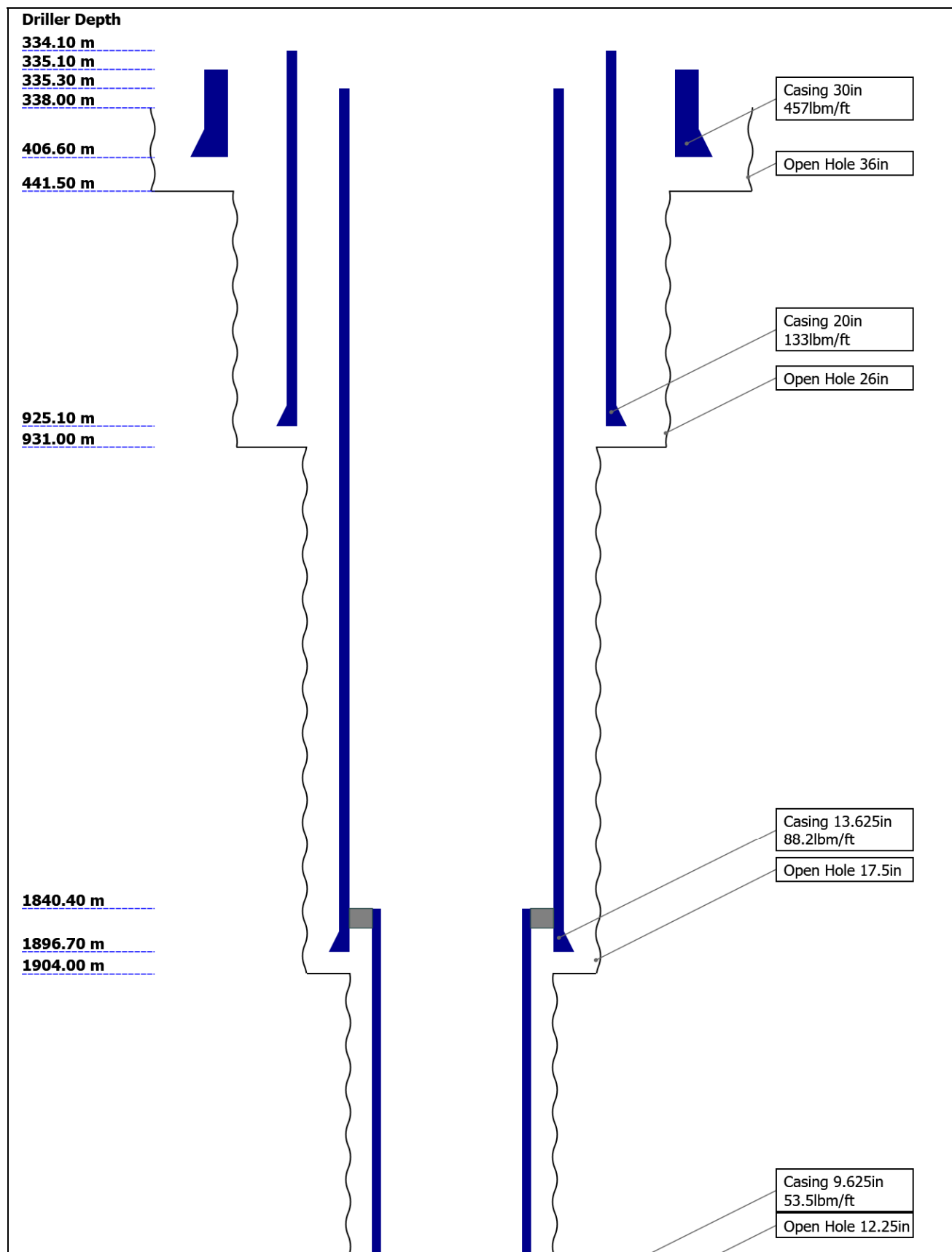
Contents

- Header
- Disclaimer
- Contents
- Well Sketch
- Borehole Size/Casing/Tubing Record
- Operational Run Summary
- Borehole Fluids
- Remarks and Equipment Summary
- Run 9 Drilling Mechanics - APWD - Time Based
 - Integration Summary
 - Composite Summary
 - Log (Drilling Mechanics Light)
- Run 10 Drilling Mechanics - APWD - Time Based
 - Integration Summary
 - Composite Summary
 - Log (Drilling Mechanics Light)
- Run 12 Drilling Mechanics - APWD - Time Based

13. Tail

- 11.1 Integration Summary
- 11.2 Composite Summary
- 11.3 Log (Drilling Mechanics Light)
- 12. Run 13 Drilling Mechanics - APWD - Time Based
 - 12.1 Integration Summary
 - 12.2 Composite Summary
 - 12.3 Log (Drilling Mechanics Light)

Well Sketch



2583.00 m
2584.00 m

2915.00 m

Open Hole 8.5in

Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2915	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	457	133	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

Operational Run Summary

Parameter (unit)	Run 9	Run 10	Run 12	Run 13		
Date Log Started	29-Dec-2019	31-Dec-2019	07-Jan-2020	13-Jan-2020		
Time Log Started	05:27:58	17:32:56	05:04:30	19:44:45		
Date Log Finished	30-Dec-2019	01-Jan-2020	07-Jan-2020	15-Jan-2020		
Time Log Finished	06:36:04	07:38:34	18:44:39	10:55:23		
Bit Size (in)	8.500	8.500	8.500	8.500		
Bit Start Depth (m)	2584.00	2596.00	2697.00	2784.00		
Bit Stop Depth (m)	2592.00	2643.00	2710.00	2915.00		
Top Log Interval (m)	2584.70	2585.90	2630.90	2699.00		
Bottom Log Interval (m)	2591.22	2642.28	2709.03	2914.28		
Max Hole Deviation (deg)	1.10	1.12	1.12	1.12		
Azimuth of Max Deviation (deg)	133.27	120.28	120.28	120.28		
Logging Unit Number	A615	A615	A615	A615		
Logging Unit Location	Service Office	Service Office	Service Office	Service Office		
Recorded By	S.Kroevel	M.Martter	M.Martter	M.Bergenkull		
Witnessed By	J.Eikenes	J.Eikenes	R.Nordstokke	R.Nordstokke		
Service Order Number	19SCA0085	19SCA0085	19SCA0085	19SCA0085		

Borehole Fluids

Remarks and Equipment Summary

APWD data from tool memory. All other data is Real Time.

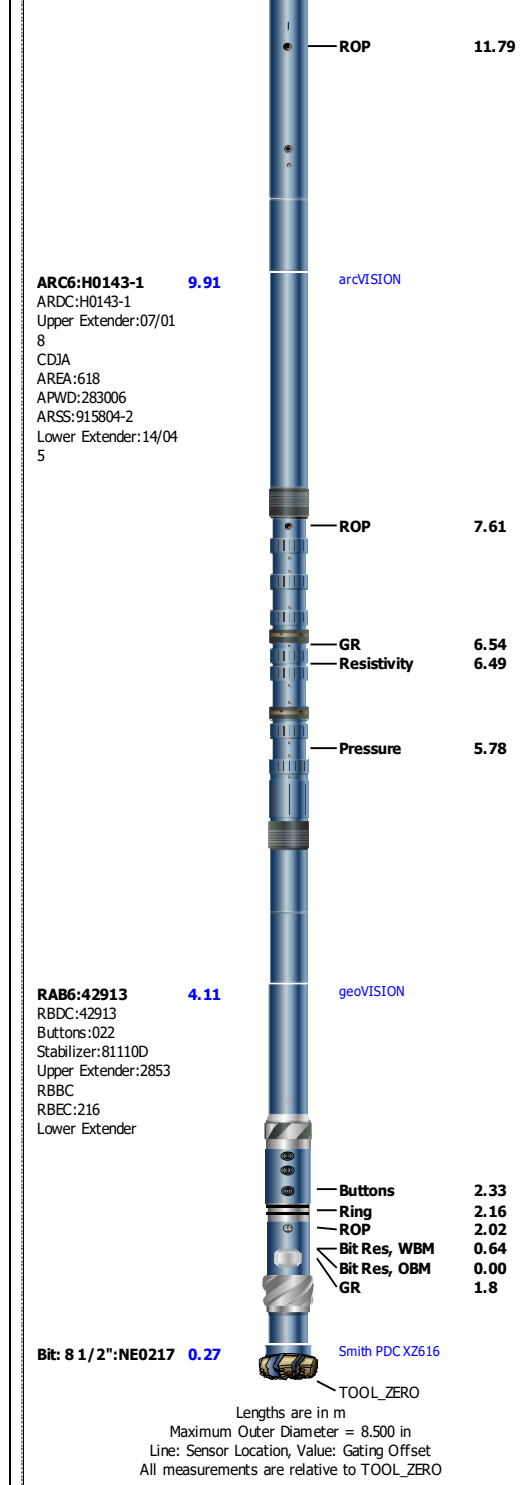
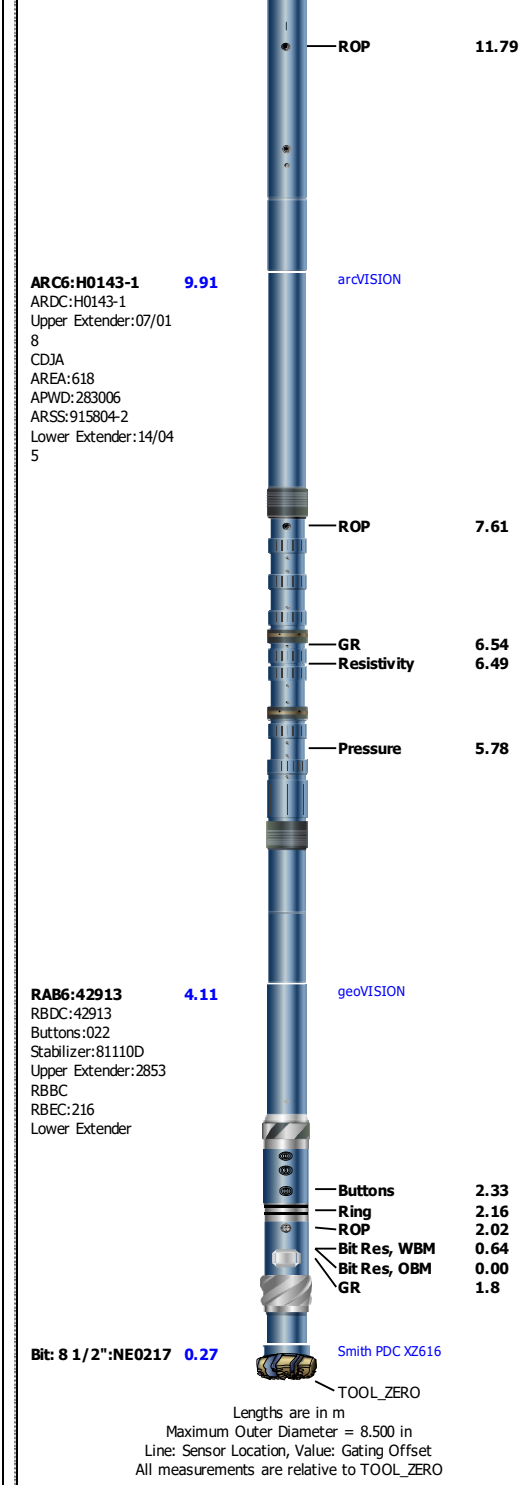
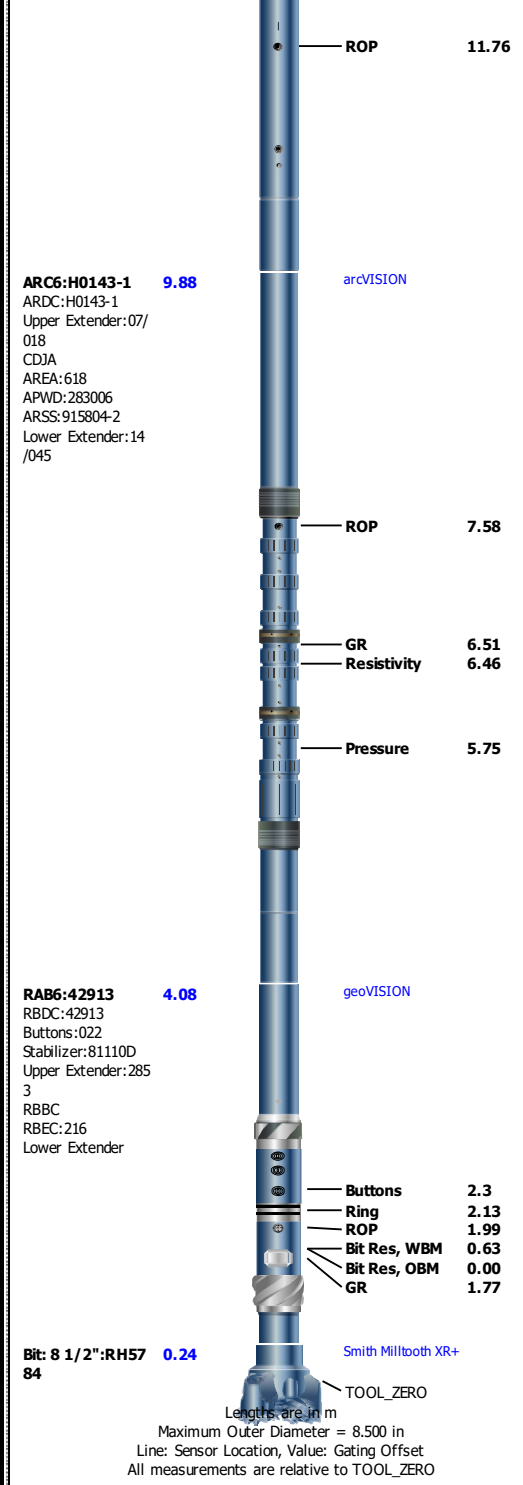
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.

Run 9: Toolstring	Run 10: Toolstring	Run 12: Toolstring
-------------------	--------------------	--------------------

Equip name	Length		MP name	Offset
TELE675:E4388	18.34		TeleScope	
MSSU675:30848				
Upper Extender				
MDC675:E4388				
MMA:2507				
MDI:3666				
PMVC:236				
PMEA:803				
MTA:2162				
MTK675:004				
MSSD675:1005482-1				
Lower Extender, Long				
		D&I		14.11
		Vibration		13.11

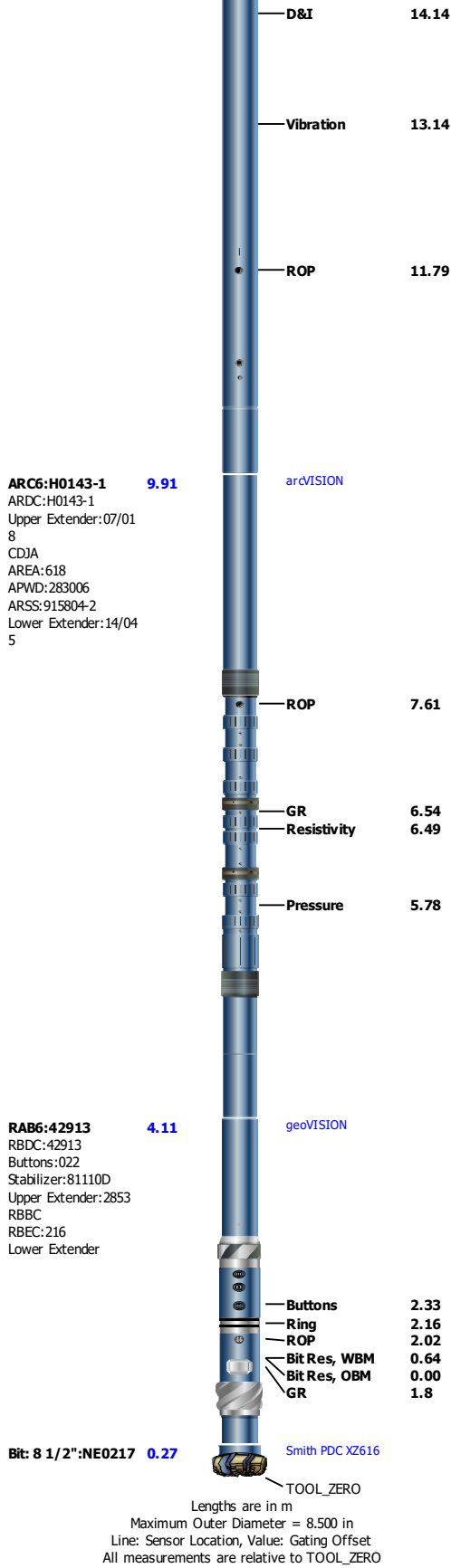
Equip name	Length		MP name	Offset
TELE675:E4388	18.37		TeleScope	
MSSU675:30848				
Upper Extender				
MDC675:E4388				
MMA:2507				
MDI:3666				
PMVC:236				
PMEA:803				
MTA:2162				
MTK675:004				
MSSD675:1005482-1				
Lower Extender, Long				
		D&I		14.14
		Vibration		13.14

Equip name	Length		MP name	Offset
TELE675:E4388	18.37		TeleScope	
MSSU675:30848				
Upper Extender				
MDC675:E4388				
MMA:2507				
MDI:3666				
PMVC:236				
PMEA:803				
MTA:2162				
MTK675:004				
MSSD675:1005482-1				
Lower Extender, Long				
		D&I		14.14
		Vibration		13.14



Run 13: Remarks
Data acquired while reaming down over cored interval from 2710 to 2784 m bit depth.
Run 13, 8 1/2 in. section and well TD at 2915 m.

Run 13: Toolstring			
Equip name	Length	MP name	Offset
TELE675:E4388	18.37	TeleScope	
MSSU675:30848			
Upper Extender			
MDC675:E4388			
MMA:2507			
MDI:3666			
PMVC:236			
PMEA:803			
MTA:2162			
MTK675:004			
MSSD675:1005482-1			
Lower Extender, Long			



Run 9

Drilling Mechanics - APWD - Time Based

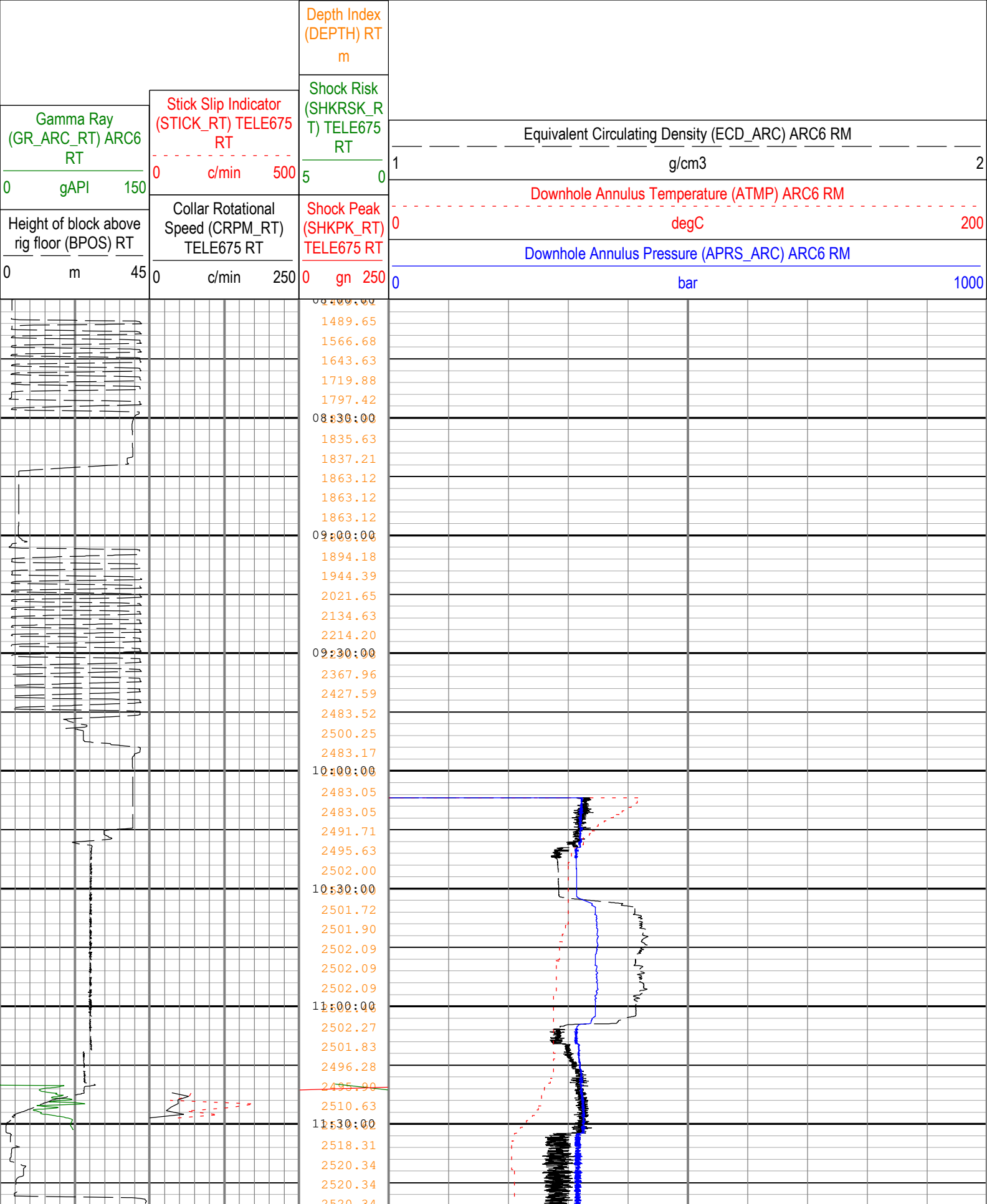
Pass Summary

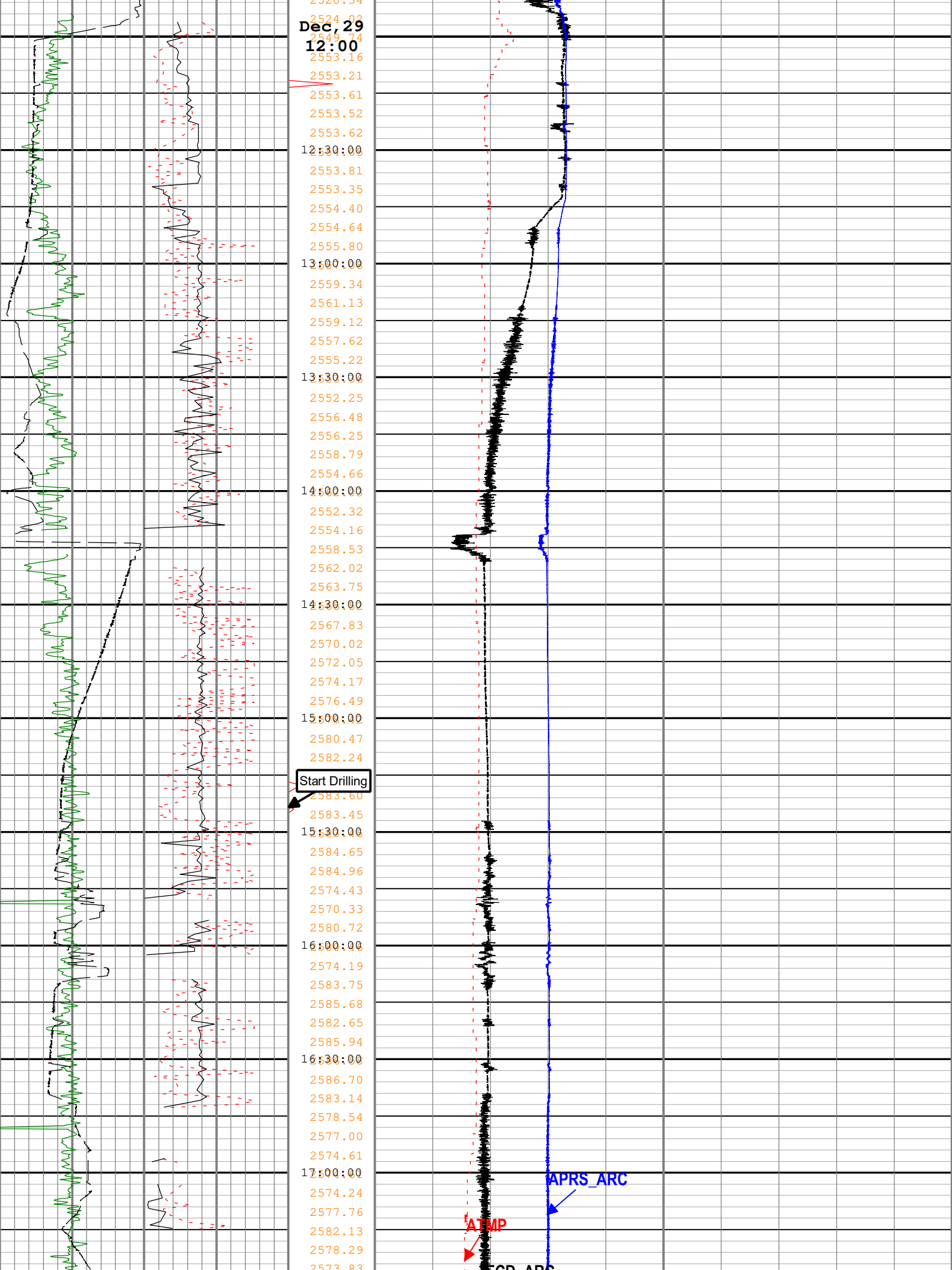
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
----------	----------------	-----------	-------	------	-----------------------

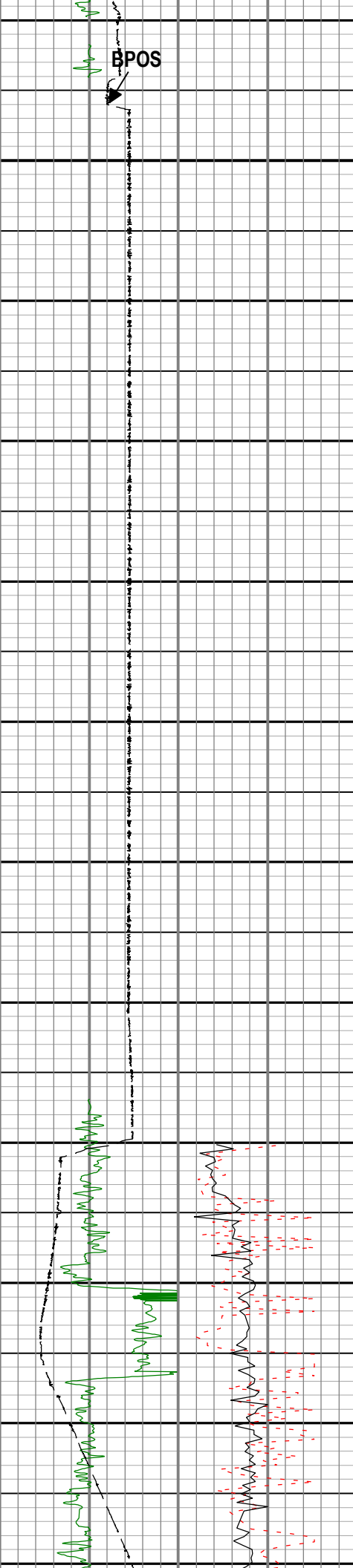
Run 9	TimeLogicalAcq	Down	29-Dec-2019 5:27:58 AM	30-Dec-2019 6:36:04 AM	Yes
-------	----------------	------	------------------------	------------------------	-----

Log	Company:Equinor Well:31/5-7				
	Run 9: TimeLogicalAcq:S054				

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:32



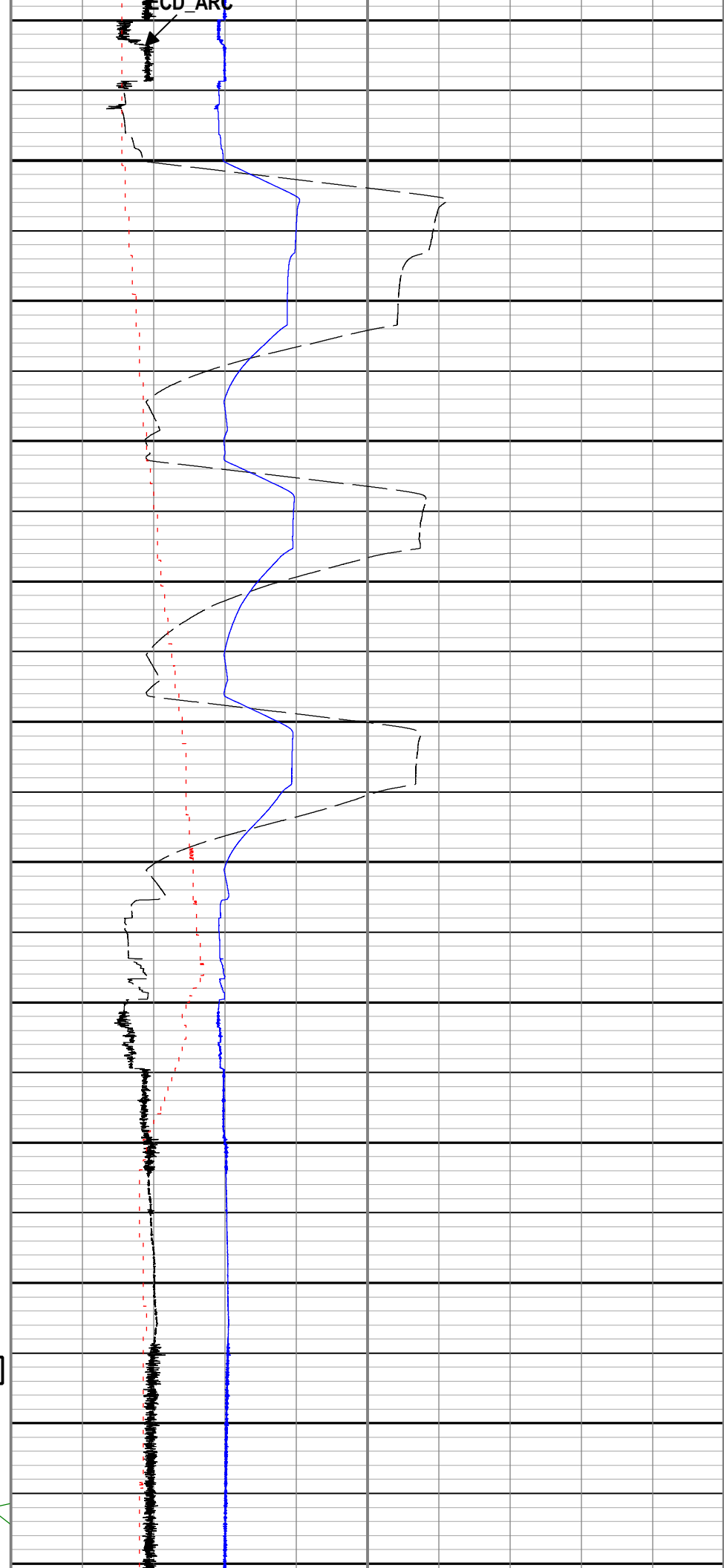


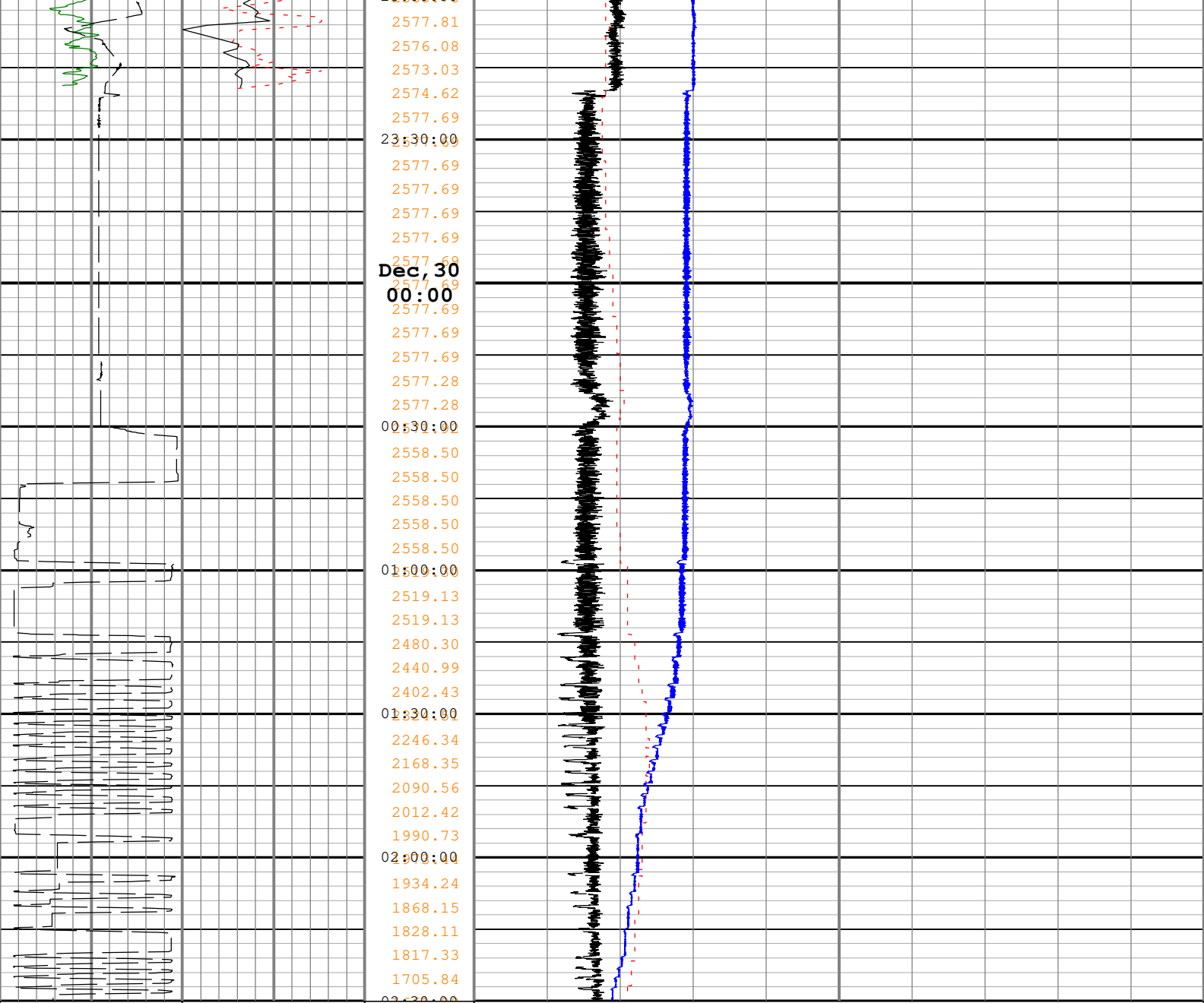


2573.03
2571.92
2571.92
2575.02
2569.41
2569.23
2569.79
2569.51
2569.33
2569.14
2569.23
2569.23
18:30:00
2569.88
2569.42
2569.23
2569.79
2569.33
19:00:00
2569.23
2569.61
2569.61
2569.33
2569.42
19:30:00
2569.70
2569.42
2569.61
2569.42
2569.61
20:00:00
2569.79
2569.14
2569.23
2569.42
2569.33
20:30:00
2569.79
2569.61
2569.14
2569.79
2569.42
21:00:00
2569.50
2569.41
2568.74
2568.74
2568.74
21:30:00
2586.68
2587.16
2587.70
2588.20
2588.97
22:00:00
2590.98
2591.96
2591.59
2586.76
22:30:00
2581.68
2578.91
2576.33
2573.75
2571.18
23:00:00

Dec, 29
18:00

Stop Drilling





Gamma Ray (GR_ARC_RT) ARC6 RT 0 gAPI 150	Stick Slip Indicator (STICK_RT) TELE675 RT 0 c/min 500	Depth Index (DEPTH_RT) RT m	Equivalent Circulating Density (ECD_ARC) ARC6 RM 1 g/cm3 2
Height of block above rig floor (BPOS) RT 0 m 45	Collar Rotational Speed (CRPM_RT) TELE675 RT 0 c/min 250	Shock Risk (SHKRSK_RT) TELE675 RT 5 0	Downhole Annulus Temperature (ATMP) ARC6 RM 0 degC 200
		Shock Peak (SHKPK_RT) TELE675 RT 0 gn 250	Downhole Annulus Pressure (APRS_ARC) ARC6 RM 0 bar 1000

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:32

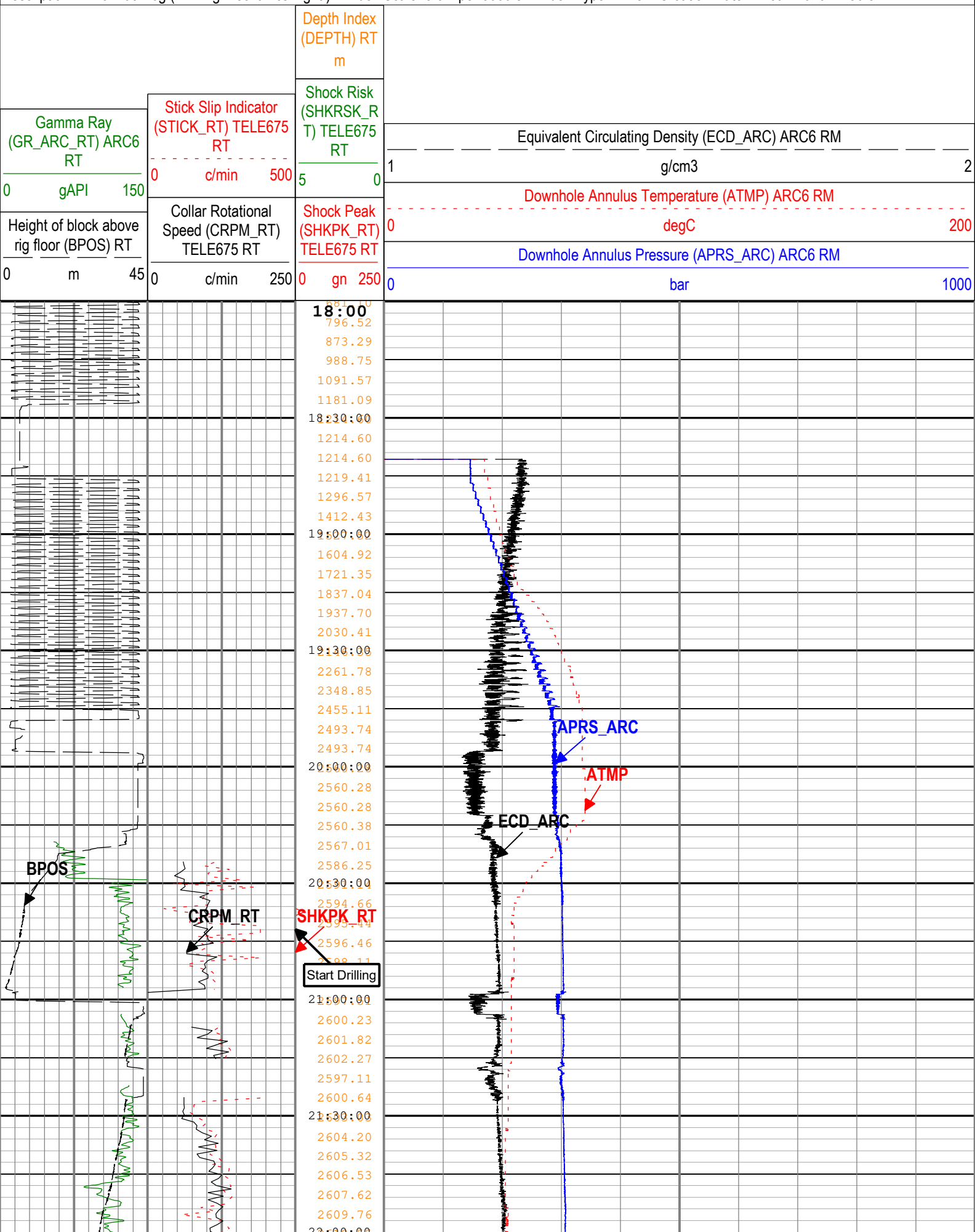
Run 10

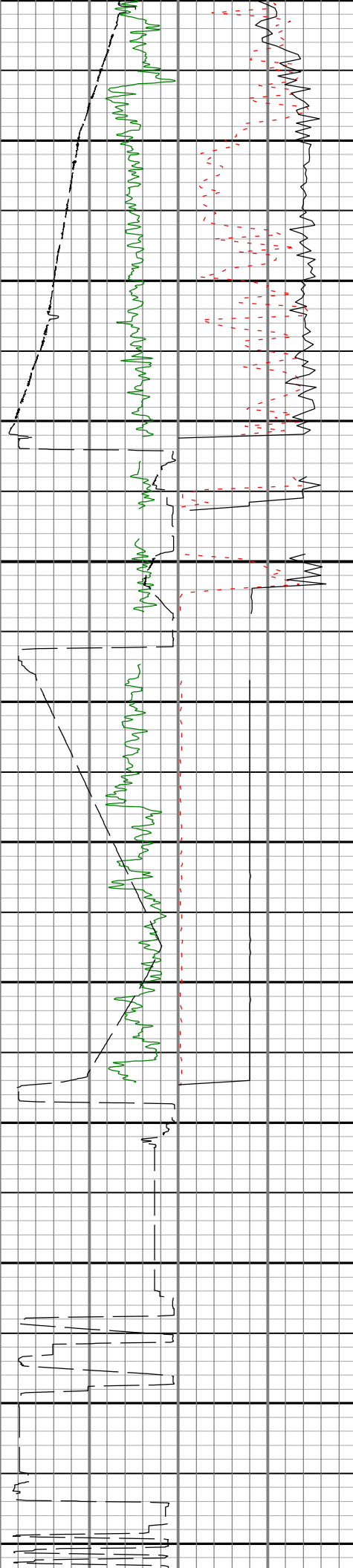
Drilling Mechanics - APWD - Time Based

Pass Summary

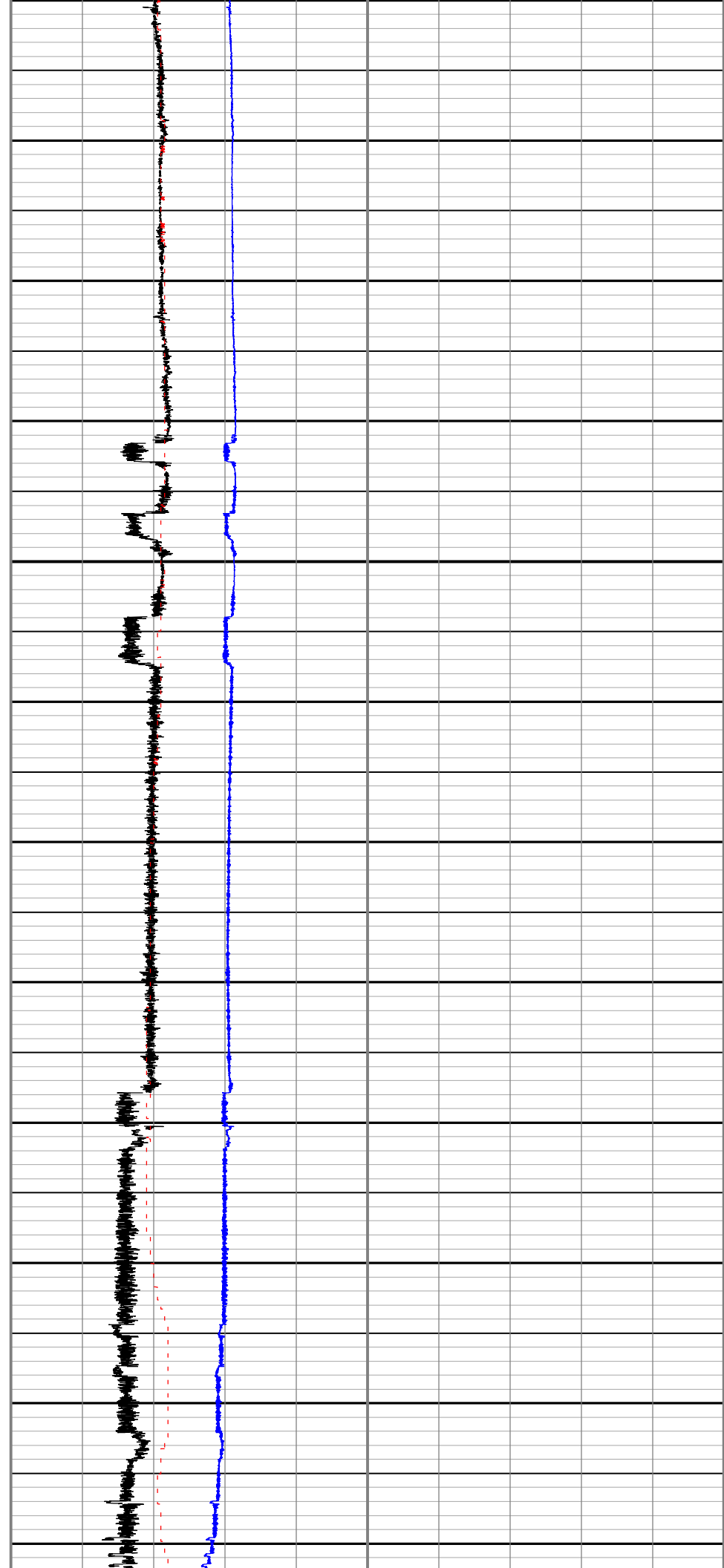
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 10	TimeLogicalAcq	Down	31-Dec-2019 5:32:56 PM	01-Jan-2020 7:38:34 AM	Yes

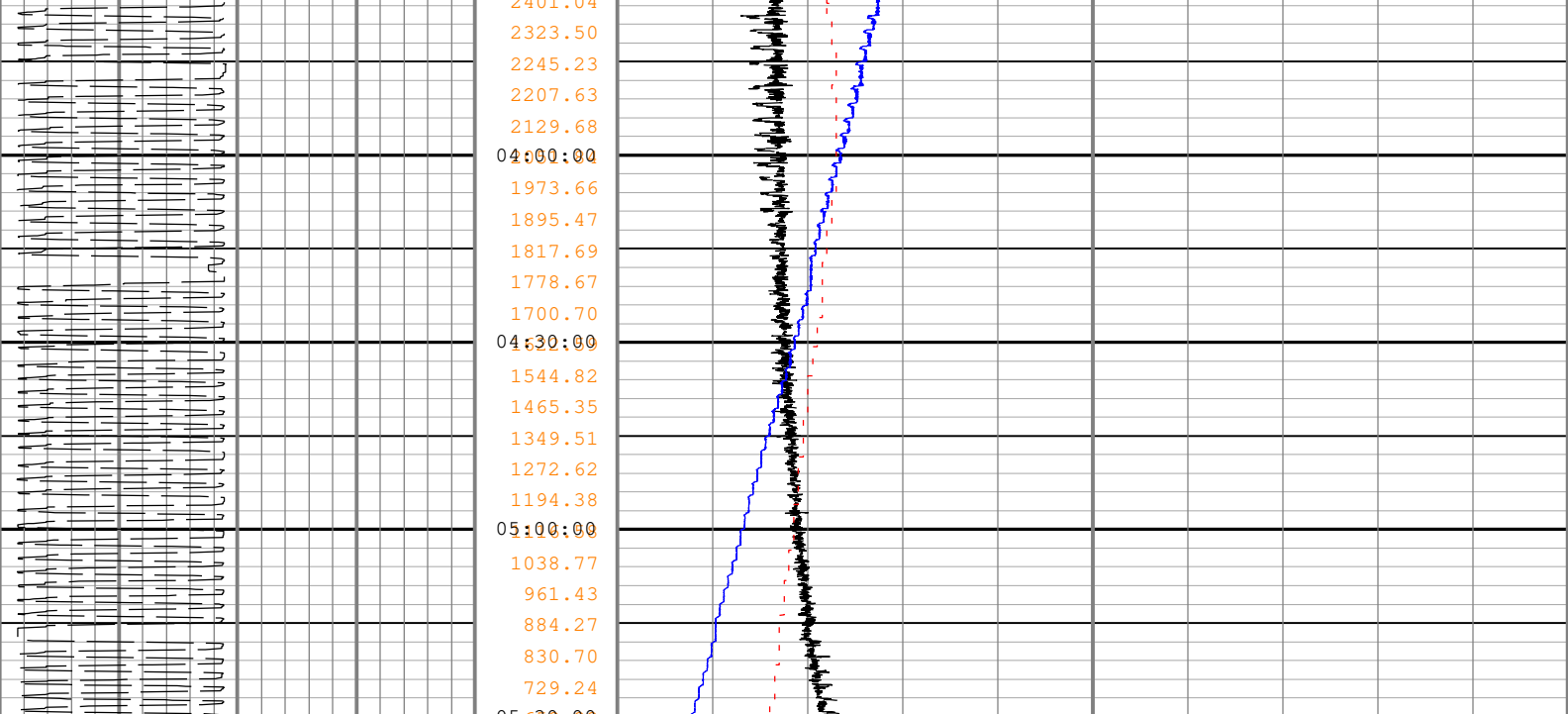
Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:34





2611.33
2613.44
2615.36
2617.25
2619.37
22:30:00
2622.08
2623.16
2624.19
2625.04
2626.52
23:00:00
2628.19
2629.11
2631.16
2632.94
2635.24
23:30:00
2636.13
2639.32
2637.74
2636.35
2636.11
Jan, 01
00:00
Stop Drilling
2635.84
2635.84
2630.94
00:30:00
2625.38
2622.61
2619.82
2617.05
2614.28
01:00:00
2608.73
2605.96
2603.19
2600.43
2601.28
01:30:00
2608.02
2611.60
2615.17
2619.61
2635.53
02:00:00
2639.85
2639.74
2639.74
2639.74
2639.74
02:30:00
2639.74
2635.10
2595.87
2595.96
2556.69
03:00:00
2556.24
2556.27
2556.33
2556.33
2517.75
03:30:00
2401.04





Gamma Ray (GR_ARC_RT) ARC6 RT		Stick Slip Indicator (STICK_RT) TELE675 RT		Depth Index (DEPTH) RT m		Equivalent Circulating Density (ECD_ARC) ARC6 RM	
0 gAPI 150		0 c/min 500		Shock Risk (SHKRSK_RT) TELE675 RT		1 g/cm3 2	
Height of block above rig floor (BPOS) RT		Collar Rotational Speed (CRPM_RT) TELE675 RT		5 0		Downhole Annulus Temperature (ATMP) ARC6 RM	
0 m 45		0 c/min 250		Shock Peak (SHKPK_RT) TELE675 RT		0 degC 200	
				0 gn 250		Downhole Annulus Pressure (APRS_ARC) ARC6 RM	
						0 bar 1000	

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:34

Run 12

Drilling Mechanics - APWD - Time Based

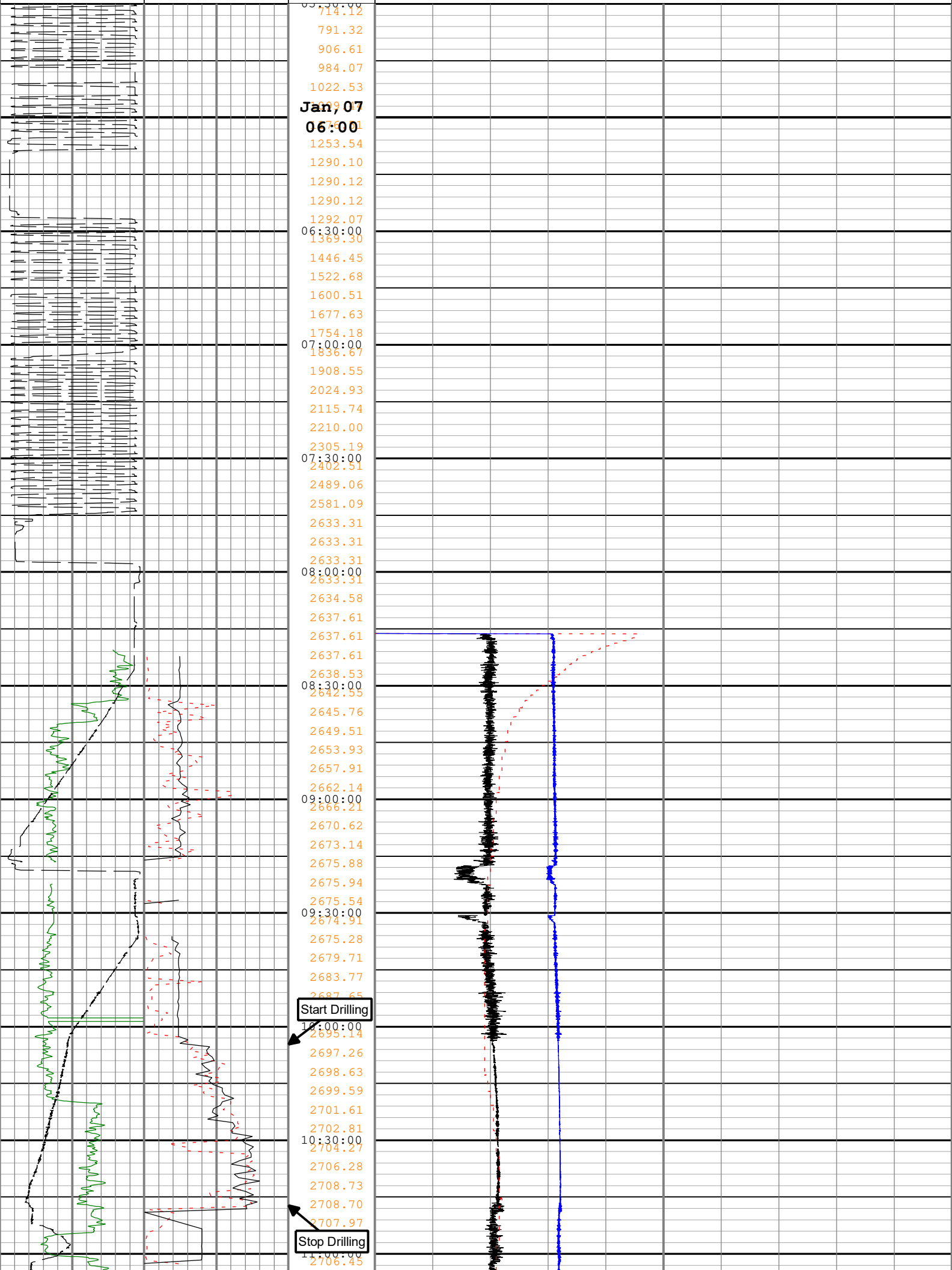
Pass Summary

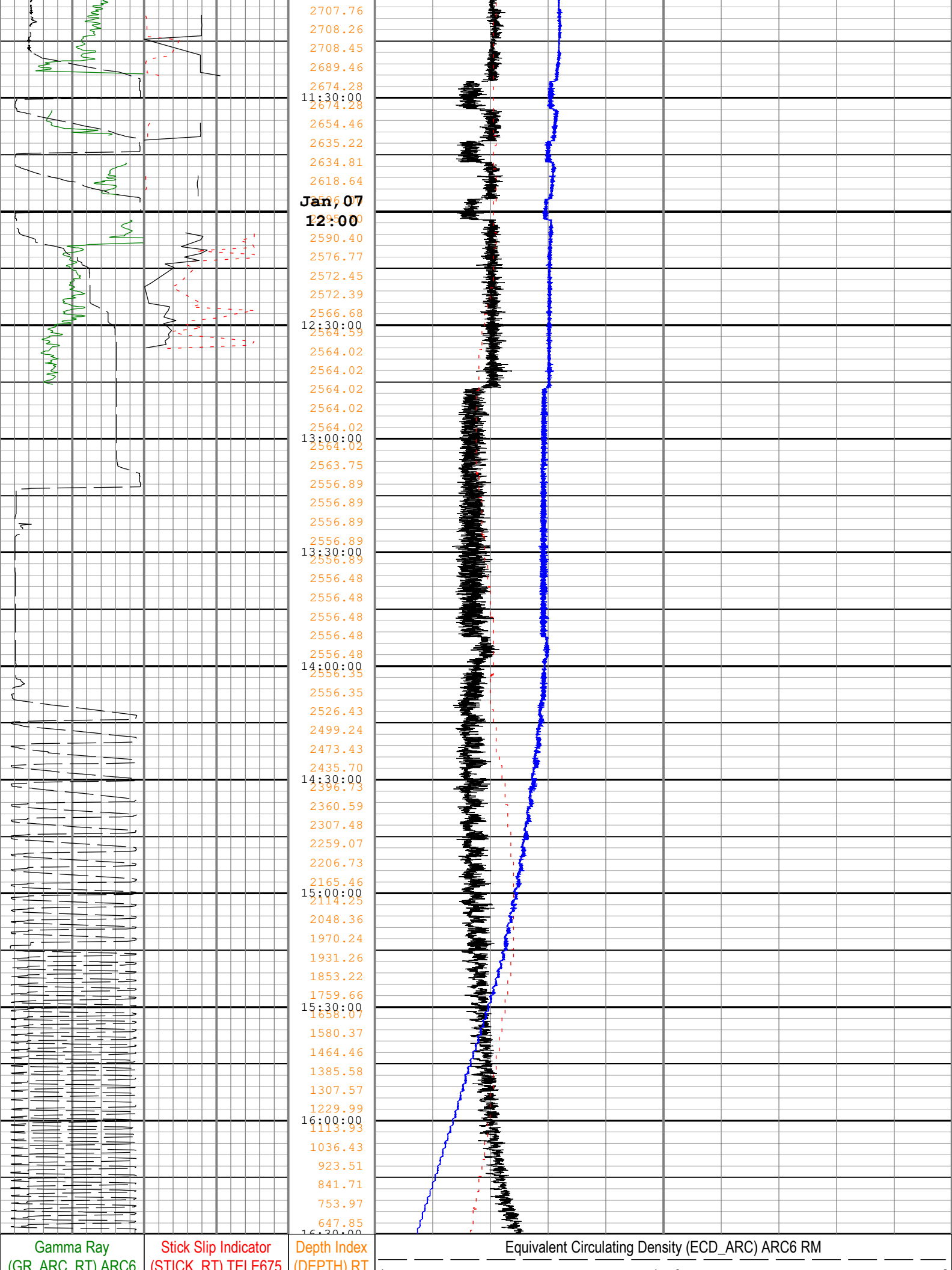
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 12	TimeLogicalAcq	Down	07-Jan-2020 5:04:30 AM	07-Jan-2020 6:44:39 PM	Yes

Log	Company:Equinor Well:31/5-7 Run 12: TimeLogicalAcq:S054
-----	--

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:36

		Depth Index (DEPTH) RT m			
		Shock Risk (SHKRSK_RT) TELE675 RT			
		5 0			
Gamma Ray (GR_ARC_RT) ARC6 RT		Stick Slip Indicator (STICK_RT) TELE675 RT		Equivalent Circulating Density (ECD_ARC) ARC6 RM	
0 gAPI 150		0 c/min 500		1 g/cm3 2	
Height of block above rig floor (BPOS) RT		Collar Rotational Speed (CRPM_RT) TELE675 RT		Downhole Annulus Temperature (ATMP) ARC6 RM	
0 m 45		0 c/min 250		0 degC 200	
		Shock Peak (SHKPK_RT) TELE675 RT		Downhole Annulus Pressure (APRS_ARC) ARC6 RM	
		0 gn 250		0 bar 1000	





RT	RT	m	g/cm3
0 gAPI 150	0 c/min 500	Shock Risk (SHKRSK_RT) TELE675 RT	Downhole Annulus Temperature (ATMP) ARC6 RM
Height of block above rig floor (BPOS) RT	Collar Rotational Speed (CRPM_RT) TELE675 RT	5 0	0 degC 200
0 m 45	0 c/min 250	Shock Peak (SHKPK_RT) TELE675 RT	Downhole Annulus Pressure (APRS_ARC) ARC6 RM
		0 gn 250	0 bar 1000

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:36

Run 13

Drilling Mechanics - APWD - Time Based

Pass Summary

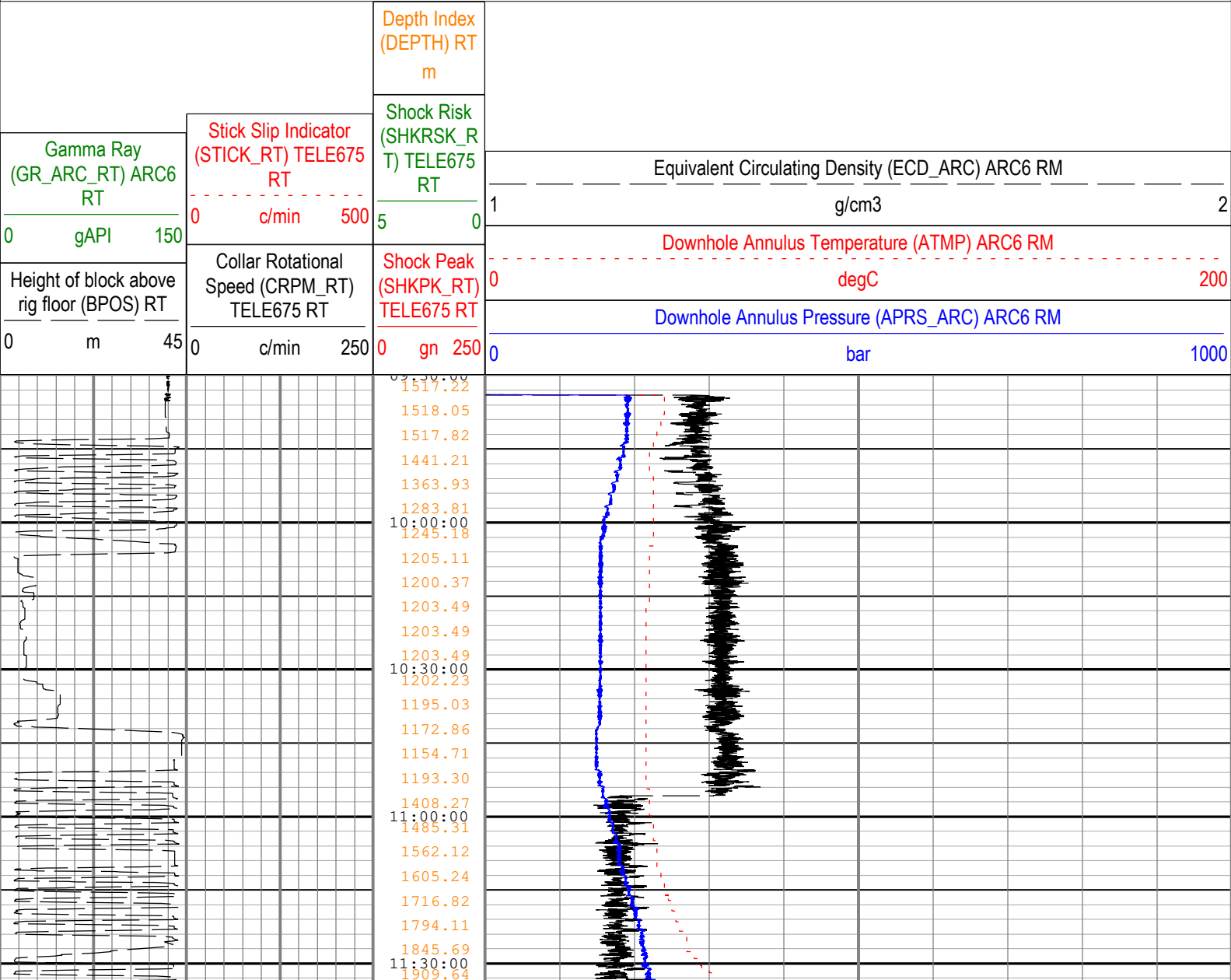
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 13	TimeLogicalAcq	Down	13-Jan-2020 7:44:45 PM	15-Jan-2020 10:55:23 AM	Yes

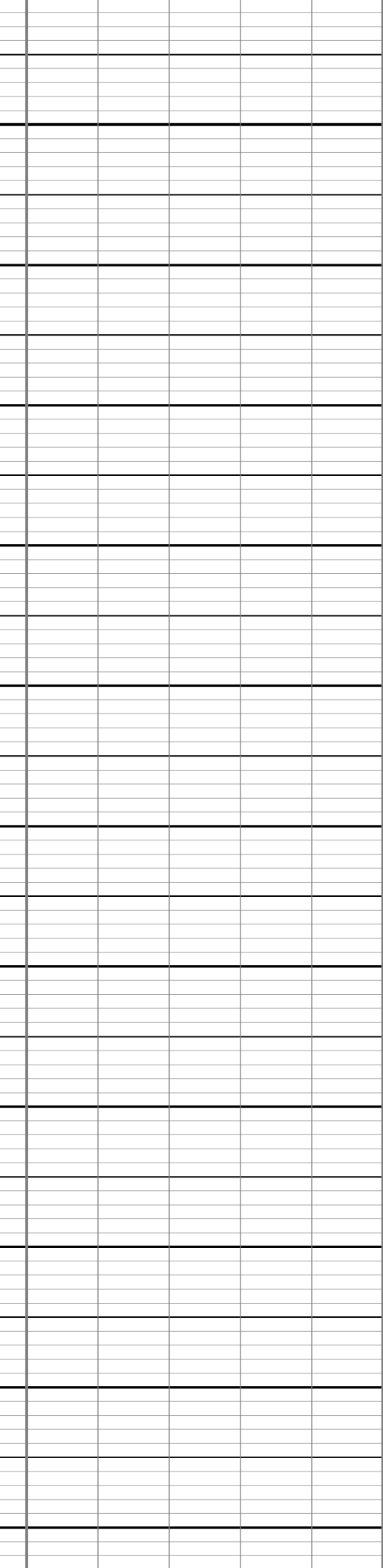
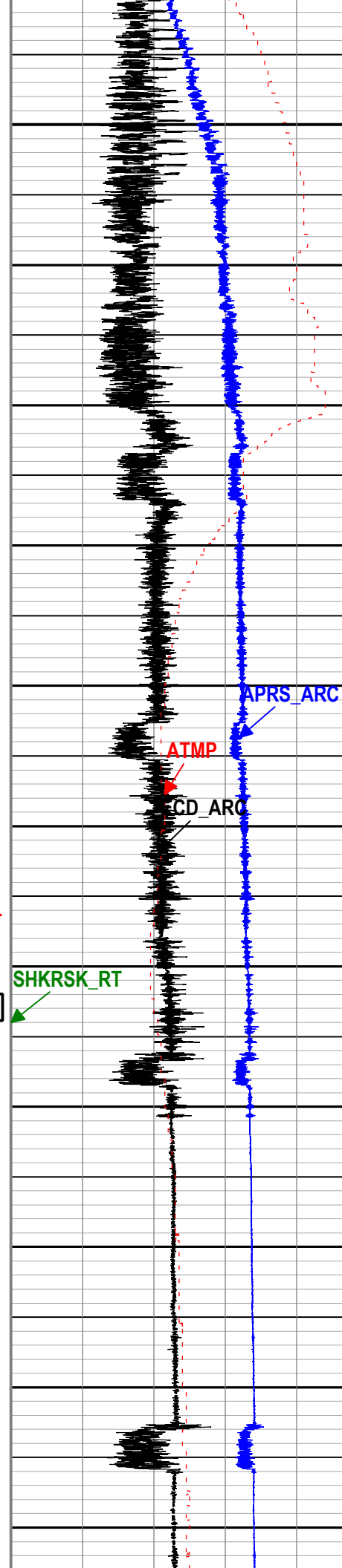
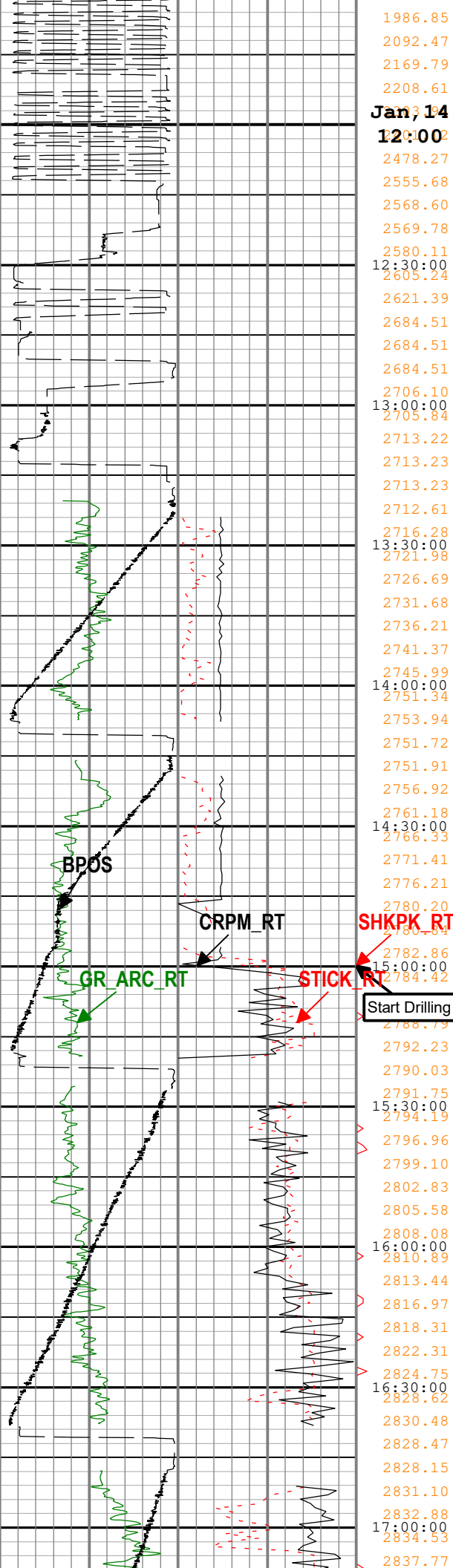
Log

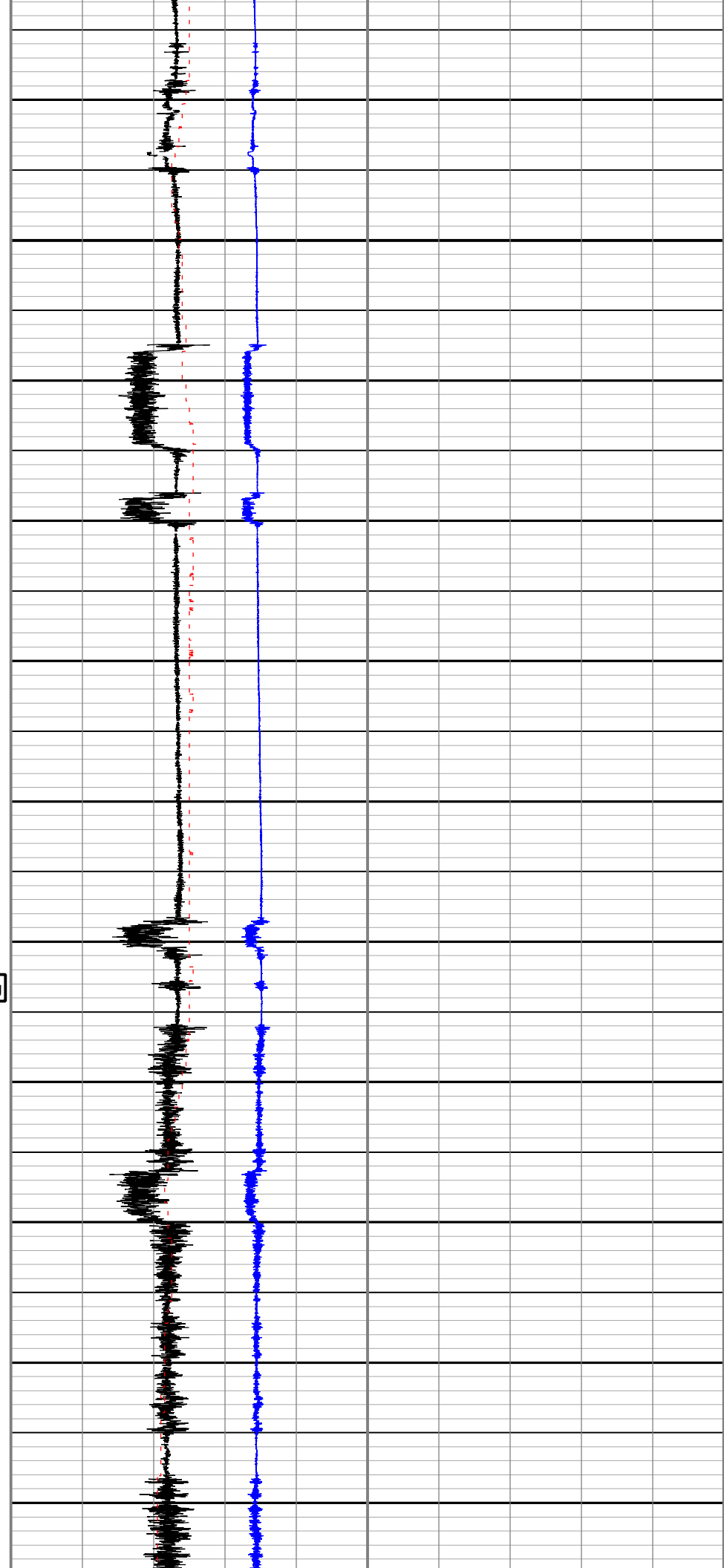
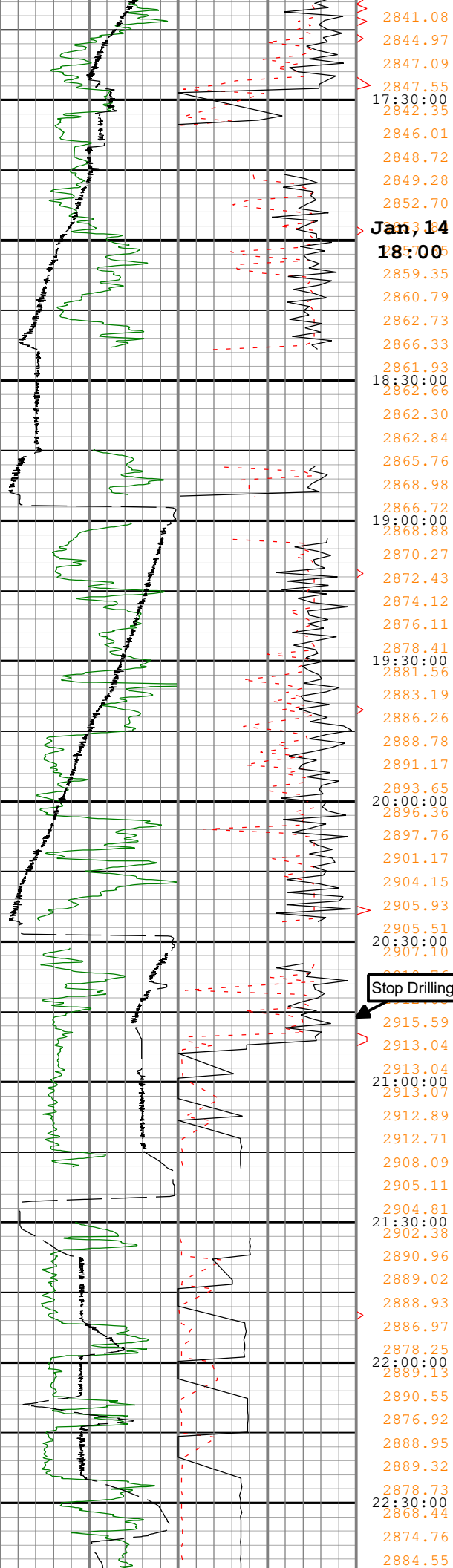
Company:Equinor Well:31/5-7

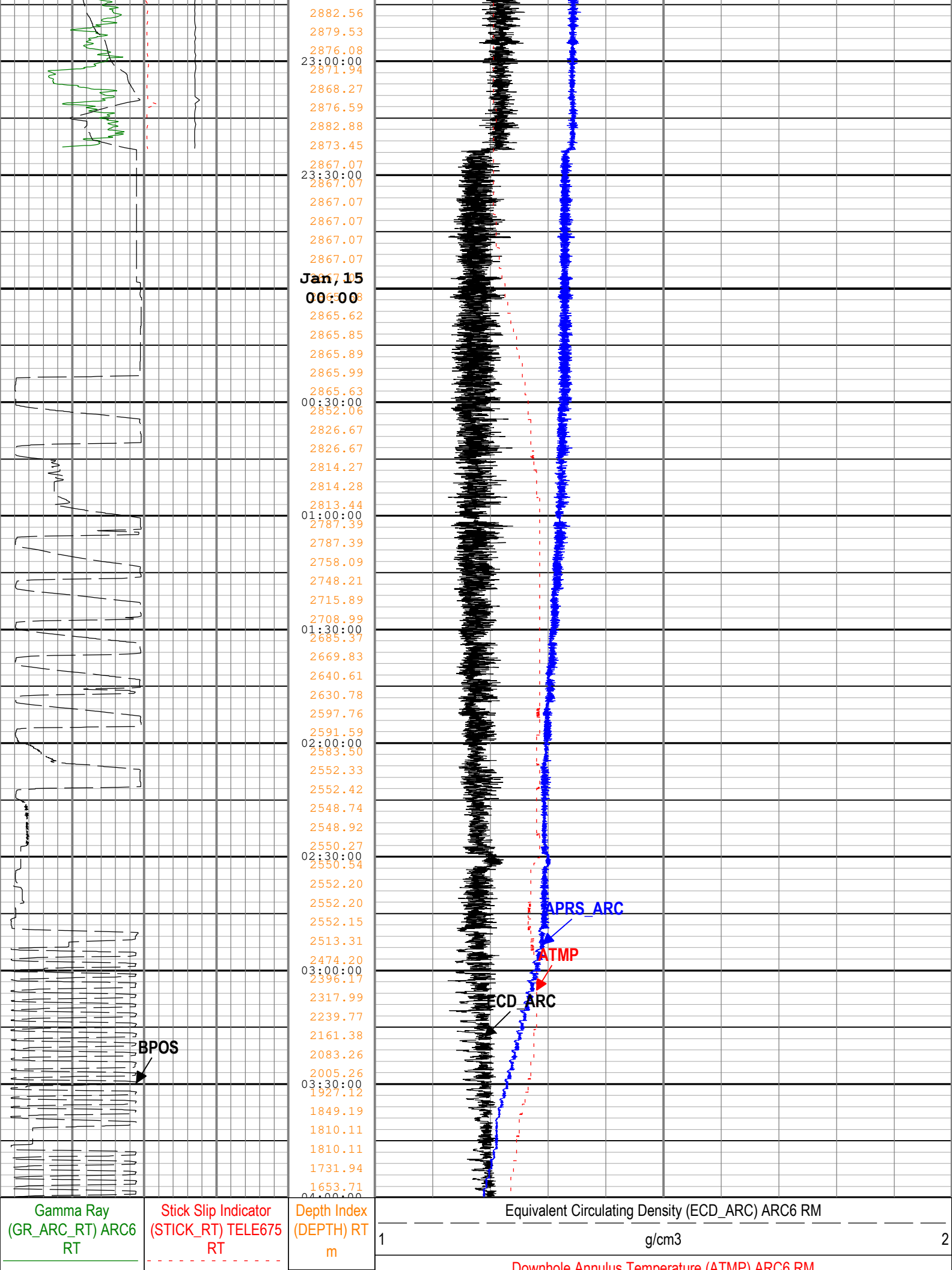
Run 13: TimeLogicalAcq:S054

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:38









0	gAPI	150	0	c/min	500	Shock Risk (SHKRSK_RT) TELE675 RT	0	Downhole Annulus Temperature (ATMR) ARC6 RM	200
Height of block above rig floor (BPOS) RT		Collar Rotational Speed (CRPM_RT) TELE675 RT					Downhole Annulus Pressure (APRS_ARC) ARC6 RM		
0	m	45	0	c/min	250	5	0	0	1000
						Shock Peak (SHKPK_RT) TELE675 RT			
						0	gn	250	
Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 12:50:38									

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



Schlumberger

Drilling Mechanics - APWD
Time Based
Real Time, Recorded Mode, Composite